

Experiment (6)

CONSERVATION OF LINEAR MOMENTUM

6.1 Introduction

In your lectures and labs and homework, you have applied Newton's Laws and the Conservation of Energy to systems which contained one or more objects. What if two or more of those objects collide with each other? Can we then apply the Conservation of Energy and Momentum?

6.2 Objectives

- Investigate the law of conservation of momentum using an isolated system of two bodies.
- Investigate the difference between elastic and inelastic collisions.
- Investigate the meaning of the coefficient of restitution and its relation to the type of collision.
- Determination of the loss in kinetic energy during inelastic collisions.

6.3 Theory

A collision occurs when the paths of two or more bodies intersect. Such a collision will cause a sudden change in the kinetic energy and in the path of each body. The magnitude and direction of the velocity of the colliding bodies may change in a collision. The force involved in a collision acts only for a very short period.

The Law of Conservation of Linear Momentum states that the total linear momentum of a colliding system of bodies is conserved if there is no external force acting on the system (i.e. isolated system).

The linear momentum of a body of mass m and velocity $\vec{v}$ is defined as,

$$\text{Linear momentum} = \vec{p} = m\vec{v} \quad (6.1)$$

which is a vector quantity.

Consider the case of a collision of two bodies along the x-axis. If the masses of the bodies are m_1 and m_2 then, according to the **law of conservation of linear momentum**, we get,

$$m_1 v_1 + m_2 v_2 = m_1 v'_1 + m_2 v'_2 \quad (6.2)$$

where v_1 and v_2 are the velocities of m_1 and m_2 before collision and v'_1 and v'_2 are the velocities of m_1 and m_2 after collision respectively.

Note that v_1, v_2, v'_1, v'_2 should be substituted in equation (6.2) with their algebraic signs, i.e., positive if the body moves in the positive x-axis and negative if it moves in the opposite direction. The choice of coordinate systems must therefore be self consistent throughout the analysis.

In general the kinetic energy before and after collision, respectively, are given by

$$K = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad (6.3)$$

$$K' = \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 v_2'^2 \quad (6.4)$$

respectively, and the loss in the kinetic energy ΔK due to the collision is given by,

$$\Delta K = K - K' = \frac{1}{2} (m_1 v_1^2 + m_2 v_2^2) - \frac{1}{2} (m_1 v_1'^2 + m_2 v_2'^2) \quad (6.5)$$

Collisions can be classified into two types:

In a Completely Elastic collision, both the momentum and kinetic energy of the system are conserved. Collisions between billiard balls are generally elastic. Collisions between subatomic particles taking place in the large Colliders are also usually elastic..

In an Inelastic collision, only momentum is conserved, but kinetic energy is not. Most of the collisions in daily life are inelastic.

An important factor to define the type of collision is called the coefficient of restitution (or the restitution factor) which is defined by,

The restitution factor =

$$-\frac{\text{relative velocity after collision}}{\text{relative velocity before collision}} = -\frac{v_2' - v_1'}{v_2 - v_1} \quad (6.6)$$

Using equations (6.2) to (6.5) and the fact $K = K'$ in the case of a completely elastic collision, we can prove the following,

- For a ***completely elastic collision***, the Restitution factor = 1
- For a ***Inelastic collision***, the Restitution factor < 1
- For a ***completely Inelastic collision***, the Restitution factor = 0, this case occurs when the two bodies stick together after collision, i.e., ($v_1' = v_2'$).

In this experiment, we will encounter two of these scenarios:

Case (1): Completely inelastic collision ($v_1' = v_2'$) and $v_2 = 0$:

Again from equations (6.2) to (6.5) we can show that,

$$p_1' = \frac{1}{1 + \frac{m_2}{m_1}} \cdot p_1 \quad (6.7)$$

Where p_1' is the momentum of first body after collision, and p_1 is the momentum before the collision.

Similarly for p_2' ,

$$p_2' = \frac{1}{1 + \frac{m_1}{m_2}} \cdot p_1 \quad (6.8)$$

If K_1' is the kinetic energy of first body after collision and K_1 is the kinetic energy of first body before the collision.

$$K_1' = \frac{1}{\left(1 + \frac{m_2}{m_1}\right)^2} \cdot K_1 \quad (6.9)$$

$$K_2' = \frac{1}{\left(1 + \frac{m_1}{m_2}\right)^2} \cdot \frac{m_1}{m_2} \cdot K_1 \quad (6.10)$$

Where K_2' is the kinetic energy of second body after collision.

Case (2): Elastic collision with $v_2 = 0$:

Again from equations (6.2) to (6.5) and $K = K'$, we can show that,

$$p_1' = -\frac{\left(1 - \frac{m_1}{m_2}\right)}{\left(1 + \frac{m_1}{m_2}\right)} \cdot p_1 \quad (6.11)$$

where p'_1 is the momentum of first body after collision, and p_1 is its momentum before the collision.

Similarly for p'_2 ,

$$p'_2 = \frac{2}{\left(1 + \frac{m_1}{m_2}\right)} \cdot p_1 \quad (6.12)$$

If K'_1 is the kinetic energy of the first body after collision and K_1 is the kinetic energy of the first body before the collision.

$$K'_2 = \frac{1}{\left(1 + \frac{m_1}{m_2}\right)^2} \cdot \frac{m_1}{m_2} \cdot K_1 \quad (6.13)$$

where K'_2 is the kinetic energy of the second body after collision.

6.4 Setup

The experimental setup and main parts are shown in Figure 6.1, Figure 6.2 and Figure 6.3.

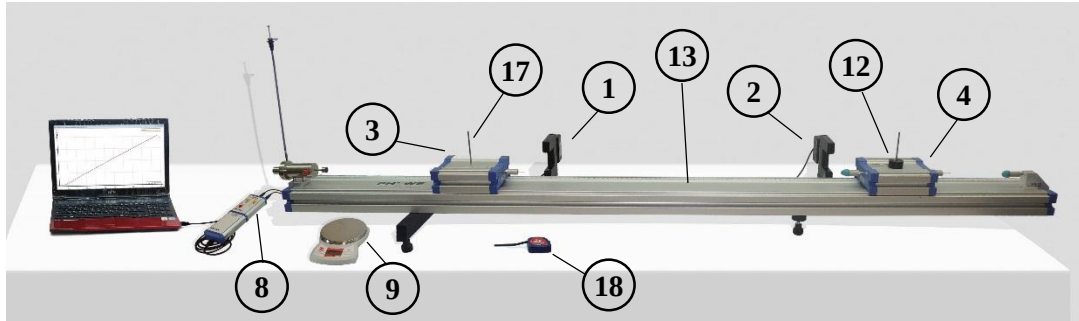


Figure 6.1: Experimental Setup.

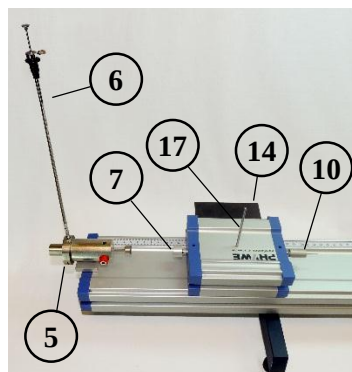


Figure 6.2: Cart 1 parts and starter system.

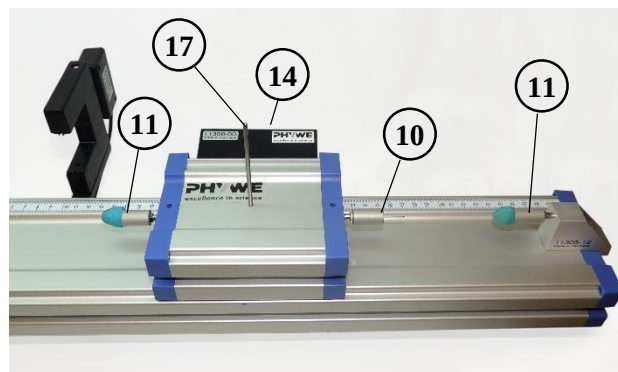


Figure 6.3: Cart 2 parts.

- 1 Light barrier 1
- 2 Light barrier 2
- 3 Cart 1
- 4 Cart 2
- 5 Starter system
- 6 Metal plunger
- 7 Holding magnet
- 8 Cobra4 sensor-unit timer/counter
- 9 Digital balance

- 10 Needle with plug
- 11 Tube with plasticine
- 12 Slotted masses
- 13 Demonstration track
- 14 Flat barrier plate
- 15 Plate with plug (Shown in Figure 6.18)
- 16 Fork with rubber band (Shown in Figure 6.18)
- 17 Metal pin
- 18 Measuring tape

6.5 Procedure

6.5.1 Task (1): Case of a completely inelastic collision ($v'_1 = v'_2$) and $v_2 = 0$.

- 1- Set up the experiment as shown in Figure 6.1 above.
- 2- Adjust the distance between the two light barriers using the measuring tape (18) to be 60 cm as shown in Figure 6.4.

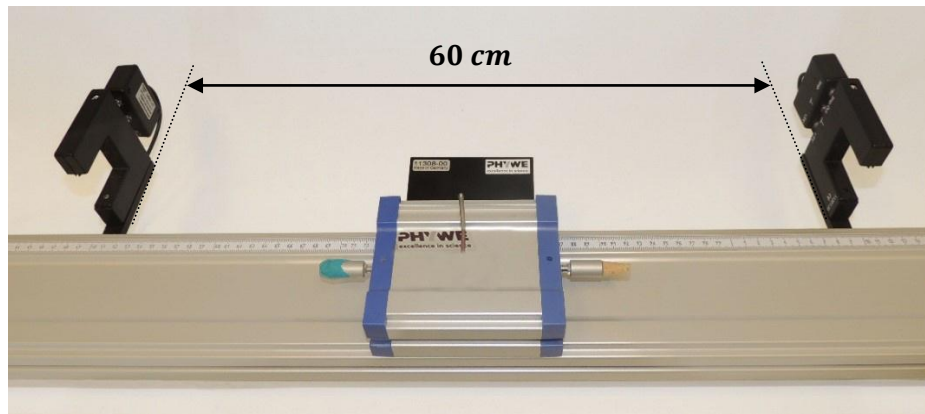


Figure 6.4: Adjusting the distance between the light barriers.

- 3- Use the digital balance (9) to measure the mass m_1 of cart 1 (3) with metal pin (17) for adding additional weights, holding magnet (7) and a needle with plug (10). Also measure the mass m_2 of cart 2 (4) with the metal pin (17), needle with plug (10) and Tube with plasticine (11).
- 4- For the first measurement, don't put extra masses on any of the carts.
- 5- On the computer desktop, click the "Measure" icon to start the software. If you have connected the USB to the computer properly, the startup screen of the Measure software will appear as shown in Figure 6.5.

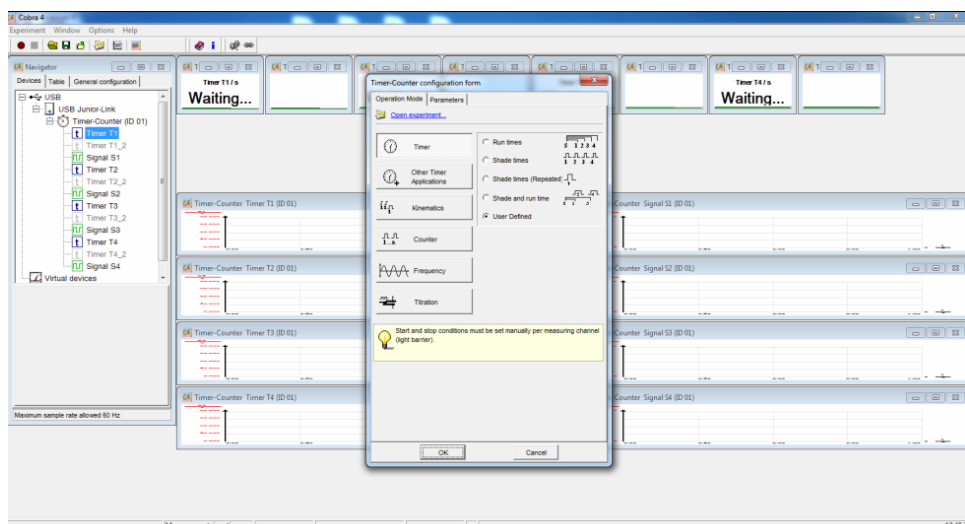


Figure 6.5: Measure Software startup screen.

6- Click “cancel”, then go to the “Experiment” menu and select “load configuration”. From the window shown in Figure 6.6, select “Desktop”, then “Law of collision” folder, and you will find a configuration file. Select it and press “OK” to load the experiment configuration. All pre-settings, which are necessary for the recording of the measured values, are now loaded.

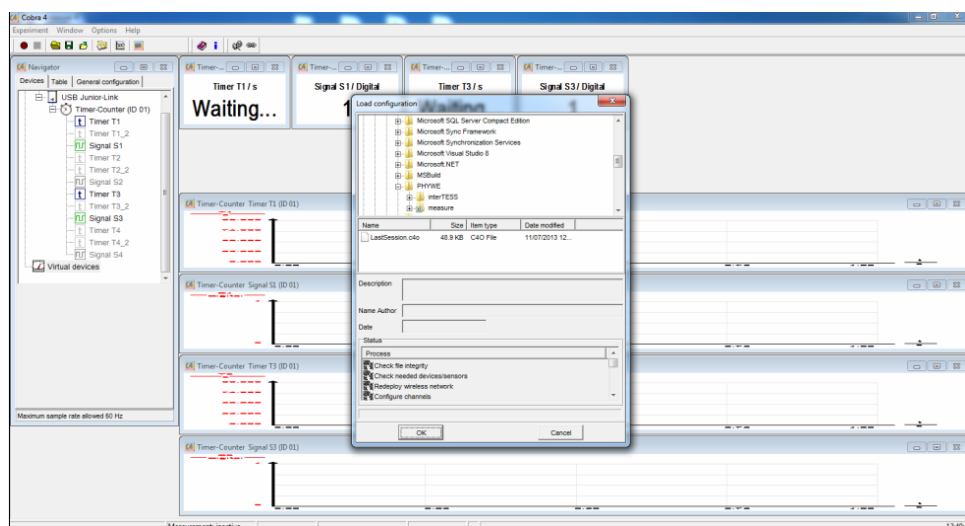


Figure 6.6: Loading the configuration.

- 7- Put cart 2 in the middle between the two light barriers as shown in Figure 6.4 above.
- 8- Set the starter system to the third position.
- 9- Attach cart 1 using the holding magnet (7) to the starter system (5) as shown in Figure 6.7.
- 10- Start recording the measurements in the Measure software from the “Record” button (●) found in the software toolbar shown in Figure 6.8.

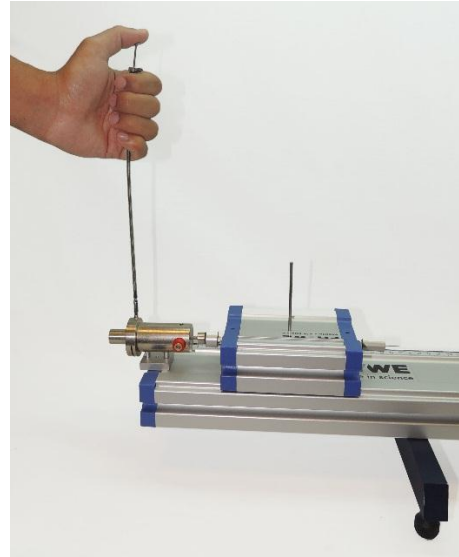


Figure 6.7: Attaching cart 1 to the starter system and pressing of metal plunger.

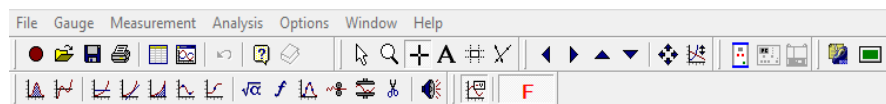


Figure 6.8: Measure software toolbars.

New windows will open as shown in Figure 6.9. where the Timer T1/s is for light barrier 1 (1) and Timer T3/s is for light barrier 2 (2).

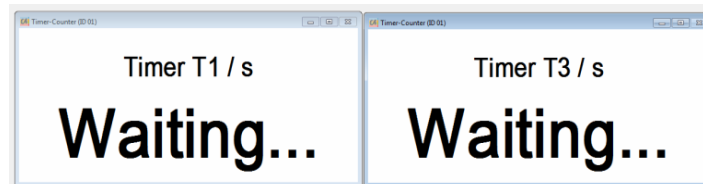


Figure 6.9: Display of Data Following Measurement

- 11- Press the metal plunger (6) of the starting system as shown in Figure 6.7 above.



The starter system gives an initial velocity to cart 1 which collides with almost the same velocity with a second cart 2 at rest. This constant velocity is achieved by the low friction in the wheels bearings and between the wheels and the track (13).



The starter system has an internal spring that can be compressed to three defined positions. You can adjust these positions.

Figure 6.10, Figure 6.11 and Figure 6.12 show the three positions of the starter system. Each of these positions will give the cart different initial velocities. The maximum initial velocity can be obtained from the third position. You can use any of these positions during this Task but it has to be fixed for all measurements.

It is recommended that the third position of the starter system is used for all measurements of Task (1).

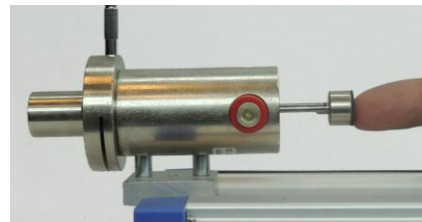


Figure 6.10: Starter system first position.

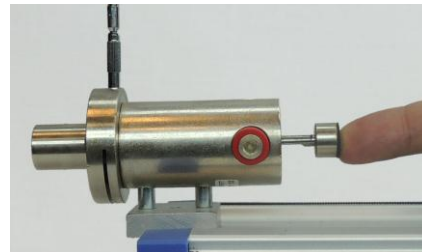


Figure 6.11: Starter system second position.

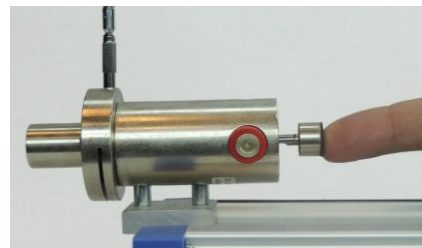


Figure 6.12: Starter system third position.



After pressing the metal plunger the inelastic collision will occur as show in Figure 6.13 and Figure 6.14.

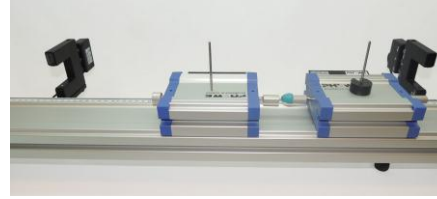
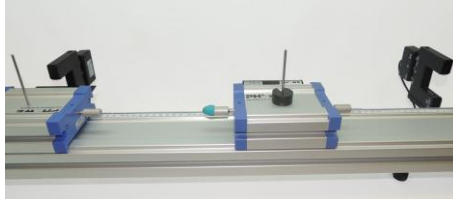


Figure 6.13: Before collision. Figure 6.14: After collision.



The light barrier measures the time during which the flat barrier attached to the car passes through it. Using the known length of the flat barrier plate = 100 mm the velocity of the cart can be determined. Figure 6.15 shows the instant in which the flat barrier plate starts cutting the light barrier beam at which point the time the counter starts counting. Figure 6.16 shows the instant in which the flat barrier plate has completely passed the light barrier beam at which the time the counter finishes counting. We divide Time by the length of the flat barrier plate to calculate the velocity and the linear momentum.

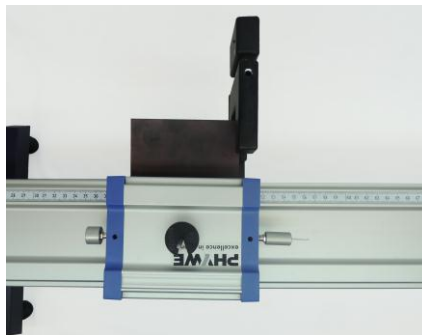


Figure 6.15: the flat barrier plate just cut the light beam (counter starts).

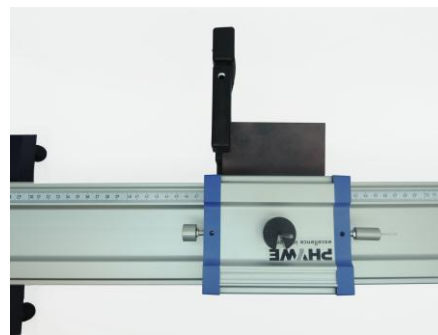


Figure 6.16: the flat barrier plate just passed the light beam (counter stops).



Never put your hands in the path of the cart, as shown in Figure 6.17. Avoid mistakenly pressing the metal plunger without preparing your Measure software. If you accidentally trigger the movement prematurely, you will have to repeat the step.

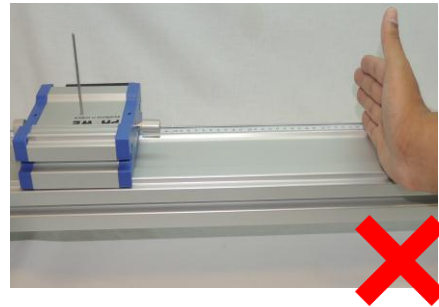


Figure 6.17: Never put your hand in the cart way.

- 12- Repeat steps 7 to 11 three more times for error analysis. For each measurement calculate the velocities of cart 1 and cart 2, the momentum of both carts before and after collision and the kinetic energies of both cart before and after collision.
- 13- Verify the conservation of linear momentum by calculating the total momentum of cart 1 and cart 2 before collision and after collision and find the error between them.
- 14- Add 400 gm mass on cart 1 so that $m_1 > m_2$ and repeat steps 7 to 11 four times for each measurement for error analysis.
- 15- Remove the extra mass from cart 1 and add 400 gm mass on cart 2 so that $m_1 < m_2$ and repeat steps 7 to 11 four times for each measurement for error analysis.
- 16- Calculate the restitution factor in each case.

6.5.2 Task (2): Case of elastic collision with $v_2 = 0$.

- 1- Remove any extra masses from cart 1 and cart 2.
- 2- Replace the needle with plug (10) of cart 1 and the tube with plasticine (11) of cart 2 with plate with plug (15) and fork with rubber band (16) respectively as shown in Figure 6.18.

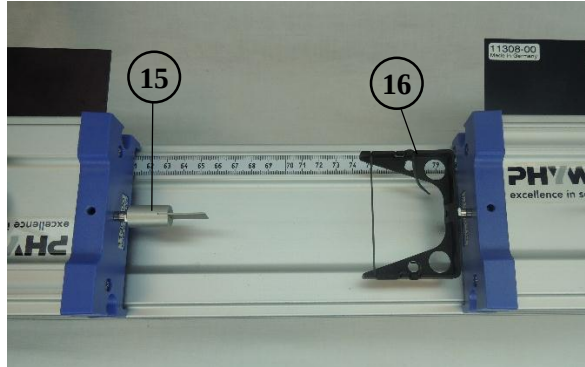


Figure 6.18: Installing the plate with plug and the fork with rubber band.



The plate with plug and the rubber band will allow an elastic collision to occur as shown in Figure 6.19 and Figure 6.20.

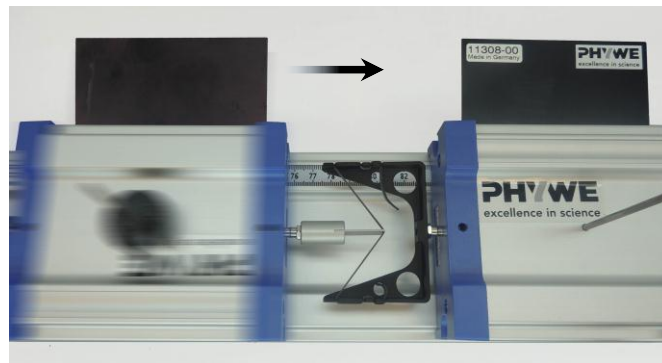


Figure 6.19: Before collision.

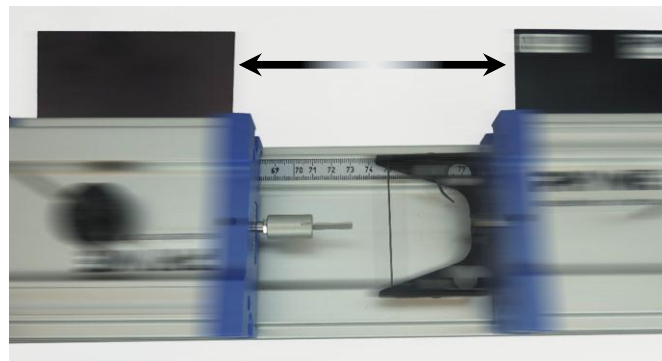


Figure 6.20: After collision.

- 3- Put cart 2 in the middle between the two light barriers.
- 4- Set the starter system to the second position.



It is recommended that the second position of the starter system is used for all measurements of Task (2).

- 5- Attach cart 1 using the holding magnet (7) to the starter system (5) as shown in Figure 6.7.
- 6- Press the “Record” button (●) to start recording the measurements.
- 7- Press the metal plunger (6) of the starter system to give cart 1 (3) an initial velocity. The cart will collide at that constant velocity with the second cart (4) which is at rest.
- 8- Repeat steps 3 to 7 three more times for error analysis. For each measurement calculate the velocities of cart 1 and cart 2, the momentum of both carts before and after collision (calculate the momentum of cart 1 after the collision by using relation (6.2) and the momentum of cart 2 before collision is zero) and the kinetic energies of both carts before and after collision.
- 9- Verify the conservation of kinetic energy by calculating the kinetic energy of cart 1 and cart 2 before collision and after collision and find the error between them.
- 10- Add 400 gm mass on cart 1 so that $m_1 > m_2$ and repeat steps 3 to 9 four times for each measurement for error analysis.
- 11- Remove the extra mass from cart 1 and add 400 gm mass on cart 2 so that $m_1 < m_2$ and repeat steps 3 to 9 four times for each measurement for error analysis.
- 12- Calculate the restitution factor in each case.